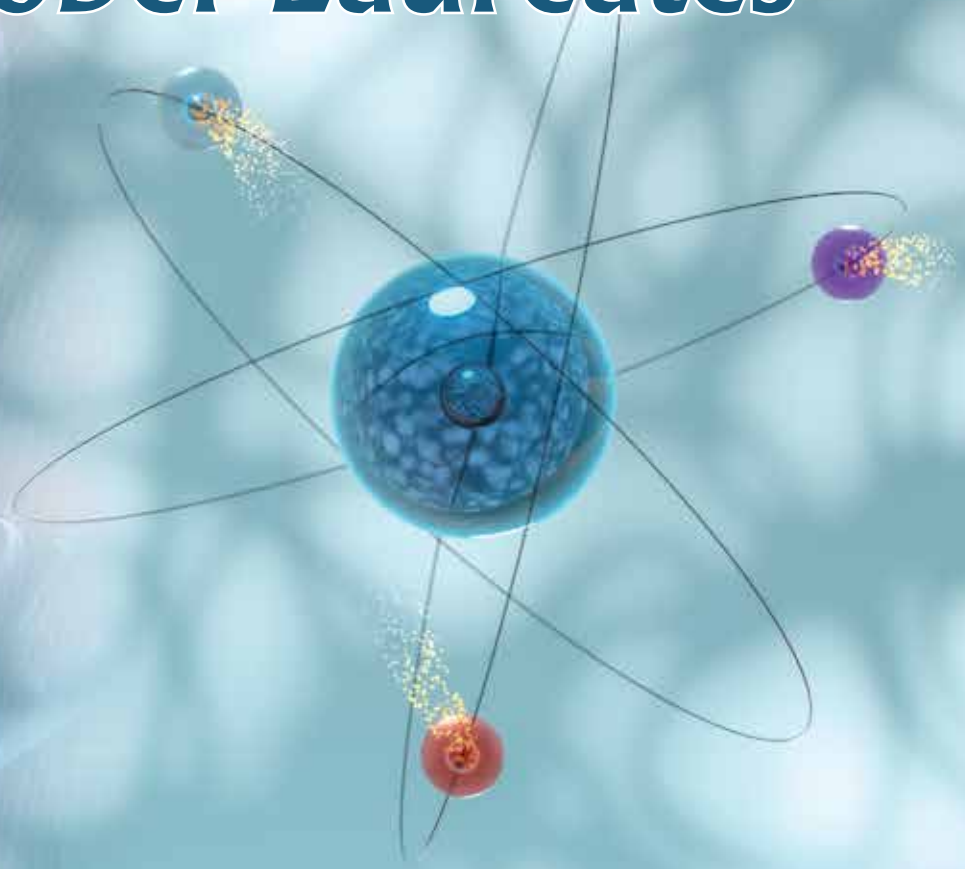
Quantum science through Nobel Laureates



Subha Das Mollick

This worksheet is suitable for classes 9, 10, 11, and 12.

Hello students! In this worksheet, we shall discuss and do some activities on the scientists who received the Nobel Prize for their work in quantum science. Do you know why we are turning our attention to quantum scientists? Because the United Nations has declared 2025 as the International Year of Quantum Science and Technology (IYQ).

Quantum computation will break all security code encryptions ... quantum cryptography will help keep our secrets ...quantum simulation will revolutionize material science and medicine ... quantum teleportation will revolutionize information protocols. If there is one buzzword that reverberates across the globe today, it is 'quantum' – all poised to take our existence to a new level.

But what is 'quantum'?
Quantum mechanics deals with the physics of the microworld – atoms, electrons, protons, and other tiny subatomic particles. Of course, since the world is built out of tiny particles, quantum mechanics also rules the macroworld around us – but in a more subtle fashion. So, the word 'quantum' really is a word that is central to our understanding of the universe.

While we talk about the scientists who won the Nobel Prize for their work in different aspects of quantum theory, you will get an idea about the core concepts that define quantum theory. Perhaps you have come across some of these concepts in your physics curriculum.

The term 'quantum' means a small packet or a bundle. The idea was first introduced in the year 1900 by a German physicist called Max Planck. He introduced this idea to explain the black body radiation curve obtained by experimental physicists. Everybody knows that a black body absorbs all energy that falls on it. It also radiates energy of all wavelengths. Max Plank had said that this absorption and emission of energy is not a continuous process. It takes place in small bundles, which he called quanta. The energy of each quantum or bundle is Planck's constant 'h' multiplied by the frequency 'ν' of the radiation. You must have come across the equation $E = h\nu$. Max Planck got the Nobel Prize for his "blackbody radiation law" in the year 1918.

The next person to take the idea of quantum forward was none other than Albert Einstein. In 1905, he was working on the photo electric effect. It had been known that when light shines on a metallic surface, electrons are ejected. But when a systematic study was done, it was found that electrons are ejected only when the frequency of light is above a certain value. So, even if infrared light shines on a metallic surface for hours, nothing will happen. But if ultraviolet light shines, immediately the electrons will escape. The lowest frequency at which electrons are ejected is called threshold frequency. In order to explain this phenomenon, Einstein said that light or any other electromagnetic energy is not only absorbed and emitted in small quanta, but light itself travels as a stream of tiny bundles of energy. He called them light quanta. And the emission of electrons is caused by the collision and exchange of momentum between light quanta and electrons.

In the year 1921, Einstein was awarded the Nobel Prize for photoelectric effect. Robert Millikan conducted his famous experiment in 1916 to prove Einstein's theory. He too was awarded the Nobel Prize in 1923.

And that is how the idea of 'quantum' progressed – through the works of theoretical and experimental physicists.

I. Albert Einstein and his Nobel Prize

In 1922, Albert Einstein was awarded the 1921 Nobel Prize in Physics "for his services to Theoretical Physics, and especially for his discovery of the law of the photoelectric effect".

We associate the formula $E = mc^2$ with Einstein. This famous formula is an expression of mass energy equivalence that comes from his special theory of relativity. But Einstein was not awarded the Nobel Prize for his mass energy equivalence formula. The year 1905 is considered *annus mirabilis* or miracle year for Einstein and for the world of Physics. Einstein was 26 years old and he was a clerk at the Swiss Patent Office. He published four groundbreaking papers this year that have revolutionized our understanding of the fundamental concepts of space, time, mass, and energy.

One of the papers was titled "On a Heuristic Viewpoint Concerning the Production and Transformation of Light". In this paper Einstein states:

> *Energy, during the propagation of a ray of light, is not continuously distributed over steadily increasing spaces, but it consists of a finite number of energy quanta localised at points in space, moving without dividing and capable of being absorbed or generated only as entities.*

Why was Einstein awarded the Nobel Prize for his photoelectric effect and not for his more famous theory of relativity? And why did the Nobel Prize come 16 years after the paper was published? Some of the answers are hidden in the award ceremony speech given by Professor S. Arrhenius, Chairman of the Nobel Committee for Physics of the Royal Swedish Academy of Sciences, You may read the entire speech following this link: https://www.nobelprize.org/prizes/physics/1921/ceremony-speech/

The bottom line is that, the Nobel Committee decided to award Einstein for his photoelectric effect because:
1. Photoelectric effect was rigorously proved experimentally by Robert Millikan in 1916.
2. Photoelectric effect had already found applications in the form of quantitative photo chemistry.

The Nobel Committee always prefers to award inventions and discoveries that benefit humanity. Einstein's photoelectric effect is one such discovery. Today, photoelectric effect has found wide applications. Your task is to figure out what these applications are, collect pictures from the internet and make a presentation. In your presentation there should be at least two to three slides explaining what photoelectric effect is.

II. List of quantum Nobel Laureates

Here is a list of some scientists who were awarded the Nobel Prize for their work in quantum science. There may be some omissions.

Year	Name of scientist	Work
1903	Henri Becquerel, Pierre Curie, Marie Curie	Nobel Prize in Physics for their discovery of natural radioactivity.
1908	Ernest Rutherford	Nobel Prize in Chemistry awarded for his work concerning the disintegration of elements and the chemistry of radioactive substances.
1911	Marie Curie	Nobel Prize in Chemistry for isolating pure radium.
1918	Max Planck	Nobel Prize in Physics in recognition of the services he rendered to the advancement of physics by his discovery of "light quanta".
1921	Albert Einstein	Nobel Prize in Physics "for his services to Theoretical Physics and especially for his discovery of the law of the photoelectric effect.
1922	Neils Bohr	Nobel Prize in Physics for his services in the investigation of the structure of atoms and of the radiation emanating from them.
1923	Robert Millikan	Nobel Prize in Physics for his work on the elementary charge of electricity and on the photoelectric effect.
1927	Arthur Compton and C.T.R. Wilson	Arthur Compton for his discovery of the effect named after him and C.T.R. Wilson for his method of making the paths of electrically charged particles visible by condensation of vapour.
1929	Louis De Broglie	Nobel Prize for Physics "for his discovery of the wave nature of electrons".
1930	C.V. Raman	Nobel Prize in Physics for his work on the scattering of light and for the discovery of the effect named after him.
1932	Werner Heisenberg	Nobel Prize in Physics for the creation of quantum mechanics, the application of which, inter allia, has led to the discovery of allotropic forms of hydrogen.
1933	Erwin Schrodinger and Paul Dirac	Dirac shared the 1933 Nobel Prize in Physics with Erwin Schrödinger for "the discovery of new productive forms of atomic theory".

1935	James Chadwick	Nobel Prize in Physics for his discovery of the neutron in 1932.
1937	Clinton Joseph Davisson	Nobel Prize in Physics for his discovery of electron diffraction in the famous Davisson-Germer experiment.
1938	Enrico Fermi	Nobel Prize in Physics to succeed in shattering even the heavier and the heaviest elements in the Periodic System. Fermi used neutrons as projectiles in his experiments.
1939	Ernest Orlando Lawrence	Nobel Prize in Physics in recognition of his invention of the Cyclotron, of its development, and of the results gained there from, especially with reference to the production of artificial radioactive elements.
1944	Otto Hahn	Nobel Prize in Chemistry in recognition of his neutron induced fission of Uranium and Thorium.
1945	Wolfgang Pauli	Nobel Prize in Physics for his "Exclusion principle".
1954	Max Born	Nobel Prize in Physics for his fundamental research in quantum mechanics, especially for his statistical interpretation of the wave function.
1965	Richard Feynman, Julian Schwinger, Shinichiro Tomanaga	Nobel Prize in Physics for fundamental work in quantum electrodynamics, with deep-ploughing consequences for the physics of elementary particles.
1967	Hans Bethe	Nobel Prize in Physics for his work on nuclear reactions.
2022	Alain Aspect, John Clauser, Anton Zeilinger	Nobel Prize in Physics for experiments with entangled photons, establishing the violation of Bell inequality and pioneering quantum information science.

- How many of the above scientists have you encountered in the film Oppenheimer?
- How many of the above scientists have worked on the Manhattan project?
- Give a prompt to ChatGPT to give you a list of Nobel Laureates in quantum science. Compare ChatGPT's list with the list above.

III. Matching scientists to their work

Below you will find two sets of thumbnail pictures. One set has pictures of scientists and the other has thumbnails of diagrams, experimental setups, graphs, etc. You will have to match the picture of the scientist with the diagram/graph associated with his work by putting the correct number in the box beside each diagram.

This is not an examination. So, take your time to do your research and fact finding on the internet.

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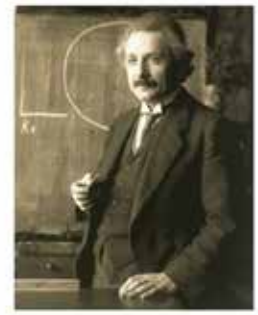
2.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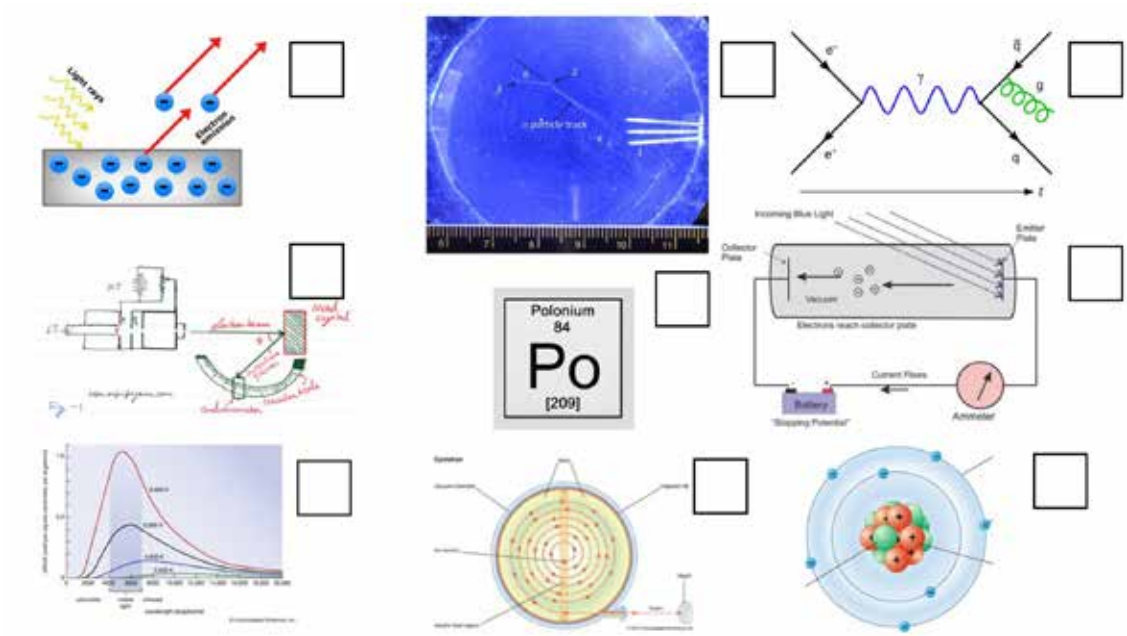

After you complete the matching successfully, create a quiz like this by selecting nine other scientists from the table in Activity II.

IV. The emerging story of the atom

In the list of Nobel Laureates is hidden the fascinating story of our emerging understanding of the atom.

- John Dalton had imagined the atom to be the smallest indivisible component of matter. He could not visualize an atom.
- Sir J.J Thompson saw that negatively charged particles are given out by matter. But matter is neutral in its charge. If negatively charged electrons come out from matter, it should also have positive charge. So he proposed the plum pudding model of the atom – negatively charged electrons embedded in a bed of positively charged matter.
- Henry Becquerel accidentally discovered that the atoms of Uranium spontaneously radiate a lot of energy – not just negatively charged particles.
- Pierre and Marie Curie discovered two new elements – Polonium and Radium.
- Rutherford's alpha scattering experiment indicated the existence of a dense positively charged nucleus at the core of the atom.
- Bohr's atomic model gave a secure position to the electrons orbiting the nucleus and explained the spectrum of hydrogen.
- James Chadwick discovered the neutron.
- Enrico Fermi used the neutron to shatter the nucleus and get new elements.
- Otto Hahn and Hans Bethe took this work forward. The atom bomb was made and deployed. The world became a dangerous place.

Your activity is to depict graphically, using sketches and diagrams, this developing idea of the atom.

V. An Indian Nobel Laureate

On the list in Activity II, you will find the name of an Indian scientist – Sir C.V. Raman. He won the Nobel Prize in 1930 for discovering the Raman Effect. C.V. Raman had done his work at the Indian Association for the Cultivation of Science, at 210 Bowbazar Street in Calcutta. Initially, he did not even have a spectroscope to take proper readings. He told G.D. Birla, the richest man in town, "Buy me a spectroscope. I will bring you the Nobel Prize in one year." G.D. Birla obliged. On February 28, 1928, Raman made public the results of his experiments. This day is celebrated as National Science Day in India.

Raman kept his word by winning the Nobel Prize in 1930. He was the first Asian to win a Nobel Prize in science. He cried at the awards ceremony because his country's flag was not there on the podium. He had to stand under the Union Jack. Raman dedicated his Nobel Prize money to the freedom fighters of India.

Here is an excerpt from the paper Raman had sent to the scientific journal *Nature*:

> *In every case in which light is scattered by the molecules in dust free liquids or gases, the diffuse radiation of the ordinary kind, having the same wavelength as the incident beam, is accompanied by a modified scattered radiation of degraded frequency.*
>
> *The new type of light scattering discovered by us, naturally requires very powerful illumination for its observation. In our experiments, a beam of sunlight was converged successively by a telescope objective of 18 cm aperture and 230 cm focal length, and by a second lens of 5 cm focal length. At the focus of the second lens was kept the scattering material, which is either a liquid carefully purified by repeated distillation, or its dust free vapour. To detect the presence of a modified scattered radiation, the method of complementary light filters was used. A blue violet filter, when coupled with a yellow green filter and placed in the incident light, completely extinguished the track of the light through the liquid or vapour. The reappearance of the track when the yellow filter is transferred to a place between it and the observer's eye, is the proof of the existence of a modified scattered radiation.*
>
> *Some sixty different common liquids have been examined in this way, and every one of them showed the effect to a greater or lesser degree. That the effect is a true scattering and not a fluorescence, is indicated in the first place by its feebleness in comparison with the ordinary scattering, and secondly by its polarization, which is, in many cases quite strong and comparable with the polarization of the ordinary scattering. The investigation is naturally much more difficult in the case of gases and vapours, owing to the excessive feebleness of the effect.*

Initially, Raman's assistant K.S. Krishnan, who had conducted most of the experiments, had called this phenomenon 'feeble flourescence'. Later, when they realized that it is not fluorescence, they called it "A new type of secondary radiation". This phenomenon is today called the Raman effect.

Elsewhere, Raman and Krishnan have written that they entered the dark lab with only a beam of sunlight coming from the window, and sat there till their eyes got used to the darkness. Then they could spot the feeble radiation coming from the liquid.

Now, your activity is threefold:

- Draw a ray diagram of the experimental setup described in the above passage.
- Make a painting of Raman's lab, showing the faint glow coming from the liquid under study. You may take the help of AI apps like Leonardo to generate this painting.
- Download pictures of Raman receiving the Nobel Prize and write a letter that Raman might have written to his wife after the ceremony.

VI. The Solvay Conference

The Solvay Conferences are organized by the International Solvay Institutes for Physics and Chemistry, founded by the Belgian industrialist Ernest Solvay. The conferences are devoted to pre-eminent unsolved problems in physics and chemistry.

Hendrik Lorentz was chairman of the first Solvay Conference on Physics, held in Brussels from 30 October to 3 November 1911. The subject was *Radiation and the Quanta*. This conference looked at the problems of having two approaches, namely classical physics and quantum theory. Albert Einstein was the second youngest physicist present (the youngest was Lindemann). Other members of the *Solvay Congress* were experts including Marie Curie, Ernest Rutherford, and Henri Poincaré. Perhaps the most famous conference was the fifth Solvay Conference on Physics, which was held from 24 to 29 October 1927. The subject was *Electrons and Photons* and the world's most notable physicists met to discuss the newly formulated quantum theory. Seventeen of the 29 attendees went on to become Nobel Prize winners.

Here are the group photos of the first and the fifth Solvay Conference:

Seated (L-R): W. Nernst, M. Brillouin, E. Solvay, H. Lorentz, E. Warburg, J. Perrin, W. Wien, M. Curie, and H. Poincaré. Standing (L-R): R. Goldschmidt, M. Planck, H. Rubens, A. Sommerfeld, F. Lindemann, M. de Broglie, M. Knudsen, F. Hasenöhrl, G. Hostelet, E. Herzen, J. H. Jeans, E. Rutherford, H. Kamerlingh Onnes, Albert Einstein and P. Langevin.
(Source Wikipedia)

A. Piccard, E. Henriot, P. Ehrenfest, E. Herzen, Th. De Donder, E. Schrödinger, J.E. Verschaffelt, W. Pauli, W. Heisenberg, R.H. Fowler, L. Brillouin; P. Debye, M. Knudsen, W.L. Bragg, H.A. Kramers, P.A.M. Dirac, A.H. Compton, L. de Broglie, M. Born, N. Bohr; I. Langmuir, M. Planck, M. Skłodowska-Curie, H.A. Lorentz, A. Einstein, P. Langevin, Ch. E. Guye, C.T.R. Wilson, O.W. Richardson
(Source Wikipedia)

- Encircle the faces of the scientists who are present in both group photos.
- In the group photo of the 5th Solvay Conference, put red arrows on the scientists who have won the Nobel Prize.
- The Conference ignited one of the most fundamental and significant debates in the history of science – known as the Bohr-Einstein debate. Watch the video "Bohr-Einstein Debate and the Copenhagen Interpretation" on YouTube and write a short paragraph on the debate, the way you have understood it. Here is the link to the video: https://youtu.be/MVEpyoVtLWk
- After you write the para in your own words, ask ChatGPT to generate a write-up on the "Bohr Einstein Debate". Underline the key words in the ChatGPT text.

VII. Artists' impressions

Artists' impressions are fundamental to the understanding of science. Galileo made copious drawings of the night sky – the moons of Jupiter, crates of the Moon and so on. Andreas Vesalius made elaborate drawings of the human anatomy in his book *De humani corporis fabrica* or *The Fabric of the Human Body*. Artists' impressions of distant places like the Andromeda Galaxy have helped us imagine the Cosmos. Artists' impressions also help in making abstract concepts a little more imaginable. Looking at sketches and cartoons, even a lay person can understand Einstein's Special Theory of Relativity to some extent.

In this activity you will make your own artistic impressions of some of the abstract concepts of quantum theory.

- Werner Heisenberg won the Nobel Prize in 1932. He is famous for his Uncertainty Principle, which states that there is inherent uncertainty in the act of measuring a variable of a particle. Commonly applied to the position and momentum of a particle, the principle states that the more precisely the position is known the more uncertain the momentum is and vice versa. Mathematically, Uncertainty Principle is expressed as:

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}$$

  Can you think of a way of expressing this mathematical formula through drawings or cartoons? The idea is that the more accurately we determine the position of a particle the more fuzzy the measurement of its velocity or momentum becomes.

- Madame Curie and her husband Pierre Curie had succeeded in extracting 100 mg of Radium chloride from one ton of pitchblende. They worked for months in a damp, dingy room with the dull grey ore gifted to them by the Austrian government. It was a stupendous task! Pierre designed experiments to study the properties of Radium. Marie had to innovate new chemical methods of extraction. The new radioactive material, if at all, was present only in traces in the pitchblende. Each step yielded crystals richer in Radium. After repeating the process a number of times, she was able to isolate the dull white powder of Radium Chloride. It looked almost like the common salt, Sodium Chloride. But it was highly radioactive – more radioactive than anything known so far. For this achievement, the couple was awarded the Nobel Prize in 1911.

  Depict in a series of three or four drawings, Pierre and Marie Curie working in their lab.

VIII. A love story that will make you sad

German chemist Otto Hahn is credited with discovering nuclear fission. He had a close collaborator called Lise Meitner who was a brilliant scientist of Jewish descent. Hahn and Meitner had discovered a new chemical element together. But Hahn later betrayed her friendship, failing to keep her from being expelled from the institute after the Nazis had gained power.

Nuclear fission is the physical process by which very large atoms like uranium split into pairs of smaller atoms; is what makes nuclear bombs and nuclear power plants possible. But for many years, physicists believed that it was energetically impossible for atoms as large as uranium (atomic mass = 235 or 238) to be split into two.

That all changed on February 11, 1939, with a letter to the editor of *Nature* – a premier international scientific journal – that described exactly how such a thing could occur and even named it fission. In that letter, physicist Lise Meitner, with the assistance of her young nephew Otto Frisch, provided a physical explanation of how nuclear fission could happen.

It was a massive leap forward in nuclear physics, but today Lise Meitner remains obscure and largely forgotten. She was excluded from the victory celebration because she was a Jewish woman. Her story is a sad one. Her name was left off scientific papers in Nazi Germany. She continued her work in Sweden. Having overcome so many obstacles in life, Dr. Meitner actively supported women in science for the rest of her career. In 1997, element 109 was named in her honour: Meitnerium.

You may watch this documentary: Lise Meitner: The Mother of Atom Bomb. Here is the link:
https://youtu.be/kbvHM2S2_bk

You may also read this blog post, which is written in a very simple language: Lise Meitner, the forgotten woman of nuclear physics who deserved a Nobel Prize. https://theconversation.com/lise-meitner-the-forgotten-woman-of-nuclear-physics-who-deserved-a-nobel-prize-106220

➢ After reading the blog post, answer these questions:

1. What is nuclear fission? How is it different from natural radioactivity?

   _____

   _____

2. How is nuclear fission a source of energy?

   _____

   _____

3. Why do you think Otto Hahn betrayed Lise Meitner?

   _____

   _____

4. In 1939, Meitner had written a letter to the editor of *Nature*, informing them of her discovery. So why did the rest of the scientific community keep quiet and not bring up her name during Otto Hahn's selection for the Nobel Prize?

   _____

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5. Download images of Lise Meitner and design an Instagram post dedicated to Lise Meitner.

IX. The Nobel Prize on entanglement

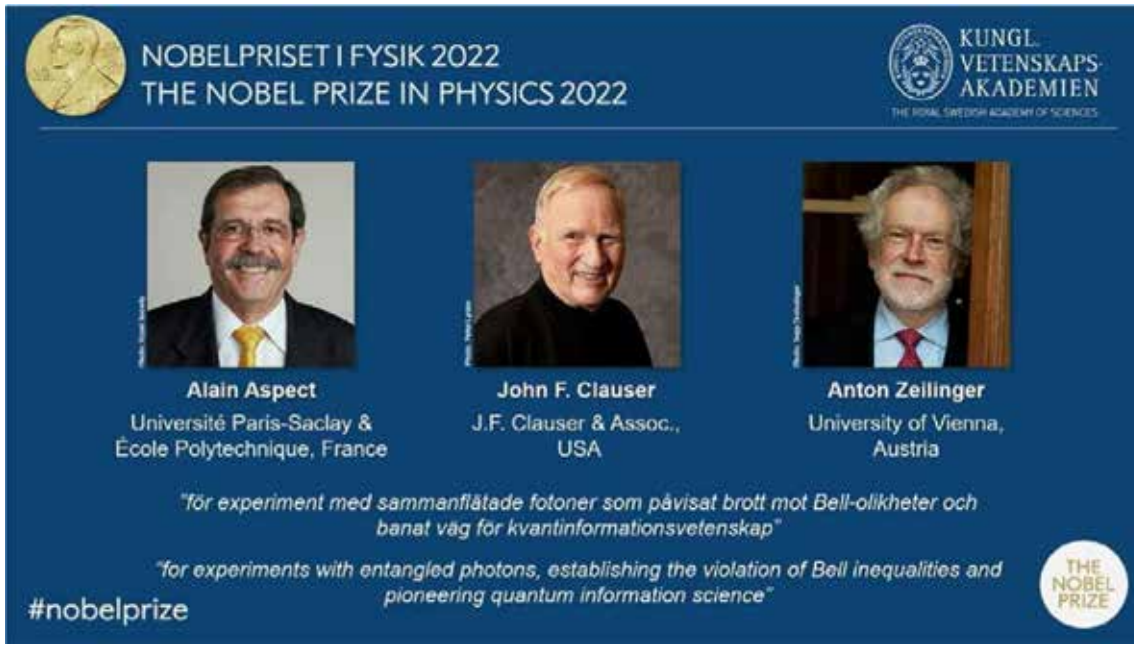
In 2022, Alain Aspect, John Clauser, Anton Zeilinger were awarded the Nobel Prize in Physics for their work with entangled photons, pioneering quantum information science. But what are entangled photons?

The idea of entanglement first came up in a paper written by Einstein, Podolsky, and Rosen. In this paper titled *Can Quantum-Mechanical Description of Physical Reality be Considered Complete?*, they proposed a thought experiment involving a pair of particles in an entangled state. That means, if the position of one particle is determined accurately, the position of the second particle could be predicted, how-so-ever far apart they are. Say for example, an electron-positron pair is created from gamma rays. And after they are created, they move apart. But since they are born from the same energy, they share the same quantum mechanical wave function and so even if they are separated far, far apart, they will remain entangled with each other.

This, in the eyes of the three scientists, was a paradoxical situation because information seemed to travel instantaneously from one particle to the other. However, Einstein's entire theory of relativity is founded on the postulate that nothing can travel faster than light. This later became known as the EPR paradox.

Einstein said that quantum mechanics was incomplete and there must be some hidden variables unknown to us. Many scientists broke their head over this paradox and tried to resolve it. One such scientist was John Stewart Bell. In 1964 he proposed another thought experiment, which later came to be known as Bell's Inequality. If Bell's Inequality is proved correct, then hidden variables do exist. If it is proved wrong, then hidden variables, as proposed by Einstein, do not exist and quantum mechanics is a complete science.

The three scientists who were awarded the Nobel Prize in 2022, proved that Bell's Inequality is wrong. This may be taken as the ultimate triumph of quantum mechanics.



Now, here are some activities for you:

- Do your own research on the internet and find out when the three scientists carried out the experiment and how they collaborated. After all, they are from three different countries.
- If 'entanglement' is a reality, then information technology will be revolutionized because information can be transferred instantaneously from one point to another. Write a short science fiction story based on the idea of entanglement.

In this worksheet we have covered almost one and a quarter century and have got some idea about how scientific thought progresses and how the work of one scientist triggers the work of another scientist. Scientists differ in their opinion, they argue and debate, and in the end science triumphs. The Bohr-Einstein debate that started at the Solvay Conference in 1927 was finally put to rest in 2022 by recognizing the work of Aspect, Clauser, and Zellinger. Einstein was proved wrong. Quantum mechanics triumphed. In the world of science, there is no father figure. Science goes only by reason. And Nobel Laureates are not the only ones who have contributed to the tremendous progress of science. There are many scientists who have not been able to join the exclusive club in spite of their ground breaking work. Let us raise a toast to the spirit of enquiry and the progress of science.

Subha Das Mollick is a filmmaker and science enthusiast. She has taught physics at the undergraduate level for more than 10 years, before switching her career to filmmaking. She has recently made a film called Dear Master, based on the relationship between Albert Einstein and Satyendranath Bose. She also has a YouTube channel and Instagram page called The Pop Steam Talk Show, where she uploads short videos on scientific themes. She can be reached at <subha.dasmollick@gmail.com>.